



人工智慧與電腦視覺研究室
Artificial Intelligence & Computer Vision Lab

多媒體創新技術研究室
Multimedia Innovative Technology Lab

智慧計算與系統研究室
Intelligent Computing and Systems Lab

BFS進階 (Segmentation)

黃祥睿 (Xiang-Rui Huang)

國立澎湖科技大學 資訊工程學系 兼任講師

Adjunct Lecturer, Department of Computer Science and Information Engineering

National Penghu University of Science and Technology Penghu, Taiwan

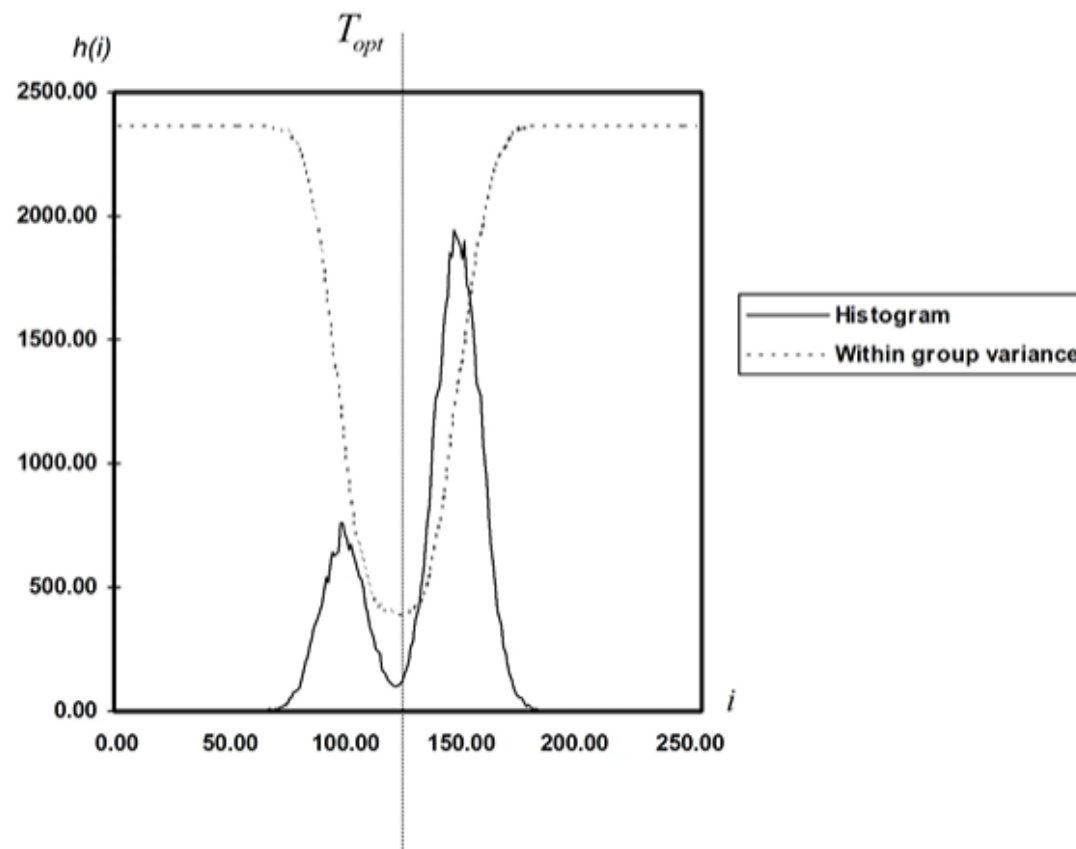
國立陽明交通大學 智慧科學暨綠能學院 博士研究生 (Ph.D. student)

College of Artificial Intelligence, National Yang Ming Chiao Tung University

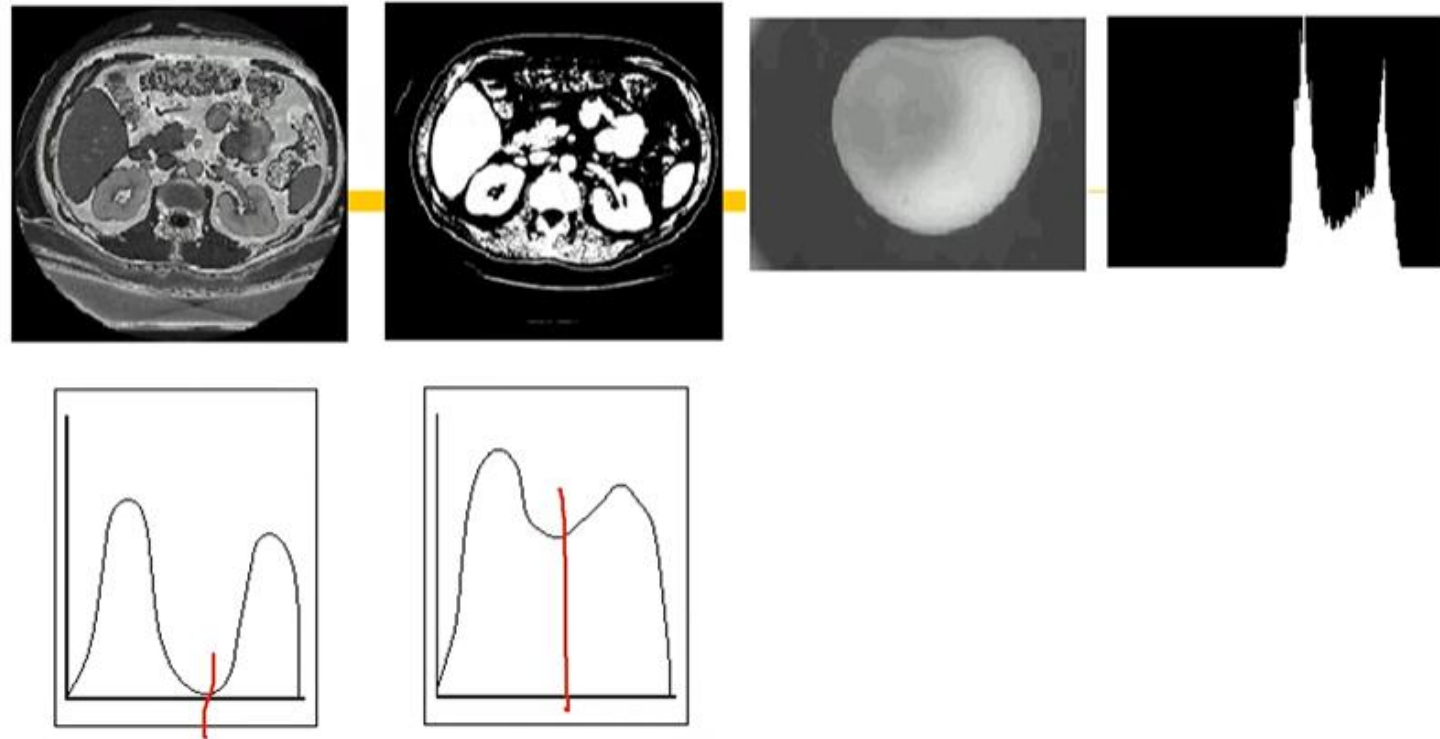


演算法

Greylevel thresholding

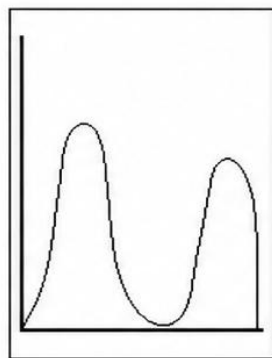


- Region Threshold using Moments



• Region Threshold using Moments

$P(i)$ = the histogram probabilities of an Image I .



$$\sigma_w^2(t) = \sigma_o^2(t) + \sigma_b^2(t)$$

where $\sigma_o^2(t) = \sum_{i=1}^t [i - u_o(t)]^2 P(i)$

$$\sigma_b^2(t) = \sum_{i=t+1}^{255} [i - u_b(t)]^2 P(i)$$

$$u_o(t) = \sum_{i=1}^t iP(i) / \sum_{i=1}^t P(i) \quad u_b(t) = \sum_{i=t+1}^{255} iP(i) / \sum_{i=t+1}^{255} iP(i)$$

$$q_o(t) = \sum_{i=1}^t P(i) \quad q_b(t) = \sum_{i=t+1}^{255} P(i)$$

$$u_o(t) = \sum_{i=1}^t iP(i) / q_o(t) \quad u_b(t) = \sum_{i=t+1}^{255} iP(i) / q_b(t)$$

Assignment 1 – BFS最佳化策略

Assignment 1: Multi-threshold

- Use the multi-threshold technique to identify the foreground and background.
 1. 建立 histogram $h(i)$
 2. 建立 cumulative sum $P(i)$
 3. **BFS 廣度優先搜尋最佳threshold 組合**
 4. 每個區間用 cumulative sum 算統計量
 5. 選出最佳 threshold 組合
 6. 根據 threshold 做影像分割
 7. 印出這一版的時間複雜度